

MATH 110C, HOMEWORK 8

DUE FRIDAY JUNE 7

Exercise 1 What is the Galois group of the polynomial $X^6 - 5$ over $\mathbb{Q}$? Over $\mathbb{R}$?

Exercise 2 Find α such that $\mathbb{Q}(\sqrt{3}, \sqrt{7}) = \mathbb{Q}(\alpha)$.

Exercise 3 Is $\mathbb{Q}(\sqrt[4]{2})$ a Galois extension of $\mathbb{Q}$?

Exercise 4 What is the Galois group of $X^4 + 2X^2 + X + 3$ over $\mathbb{Q}$?

Exercise 5 Do Problems 4 and 5 in Exercises 55.9 p.336 and Problem 6 in Exercises 57.13, pp.349–351.

Exercise 6 (Corrected version of Problem 7 in Exercises 57.13).

Let L/F be a finite Galois extension, $L/K_i/F$ intermediate fields, and $H_i = G(L/K_i)$ for $i = 1, 2$. Show $H_1 \cap H_2 = G(L/K_1(K_2))$ and the fixed field of the smallest group in $G(L/F)$ containing H_1 and H_2 is $K_1 \cap K_2$.